Multiphoton interference and representation theory in the many-photon limit

Doctoral Candidacy Exam

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Bogotá, Colombia
2026

Introduction

Calculating multiphoton scattering probabilities in linear optical networks presents analytical and computational challenges that increase with the number of photons and with the inclusion of partial indistinguishability [1, 2]. This problem is relevant to quantum technologies such as boson sampling and is closely related to computational complexity [3, 4, 5]. Recent studies have established semiclassical correspondences in the many-photon limit [6, 7]. Building on these results, this proposal addresses three questions: How is the asymptotic behavior of multiphoton probabilities related to classical interferometry? How do the irreducible sectors of $U(M)$ contribute to these probabilities when photons are partially indistinguishable? How can a quantitative correspondence be established between the semiclassical behavior of the matrix elements of these representations and the classical map from relative phases to intensities?

This research focuses on the many-photon limit, $N \to \infty$, with a fixed number of modes M and sequences of occupations whose fractions approach fixed values. As illustrated in Figure 1, this proposal addresses the problem by connecting three domains that may appear distinct at first: quantum multiphoton interference, the classical map associated with classical multiport interferometry, and the theory of irreducible representations of $U(M)$ in the limit of large Young diagrams.

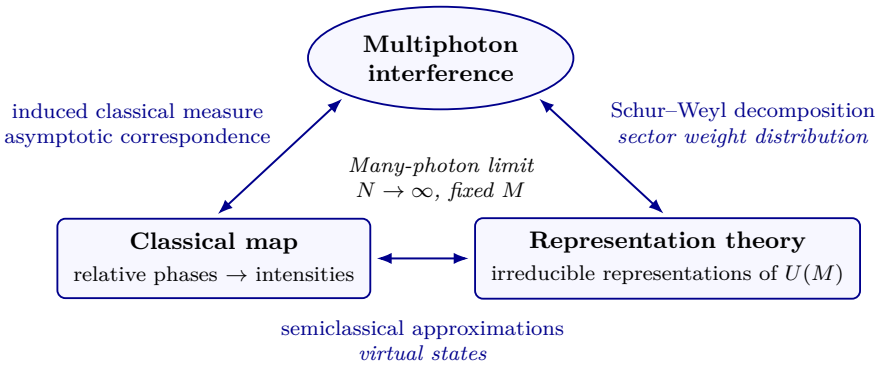


Figure 1: Correspondences between multiphoton interference, the classical map from relative phases to intensities, and the representation theory of $U(M)$ in the asymptotic multiphoton limit.

The first correspondence relates multiphoton probabilities to classical interferometry. In the many-photon limit, the macroscopic behavior of these probabilities is related to a classical measure induced by a map from relative phases to intensities [6, 7]. In our previous work, we expressed multiphoton states exactly through an expansion in tensor powers of auxiliary single-photon states, referred to as virtual states [7]. These states are parametrized by the relative input phases, and their propagation through the interferometer enables the definition of the classical map $C_{\rho,U}(\boldsymbol{\phi})$, which maps the torus of relative phases $\boldsymbol{\phi} \in \mathbb{T}^{M-1}$ to the simplex of normalized output intensities Δ_{M-1}. The probabilities are exponentially suppressed at a positive distance from the image of the map, and, at regular points, their rescaled slowly varying component converges to the density of the induced classical measure. Under additional conditions on the interferometer and the rank of the effective matrix, this convergence also holds for the rescaled physical probabilities. When $M \geq 3$, the map may exhibit caustics and interior voids, the latter corresponding to regions without preimages. A general characterization of the interference contributions associated with different preimages, especially near caustics, remains one of the problems motivating this research.

The second correspondence relates multiphoton interference to representation theory. Schur–Weyl duality provides an exact decomposition of the spatial state space into sectors $\mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda$, where $\mathcal{H}_\Lambda$ carries an irreducible representation of $U(M)$ labeled by a Young diagram and $\mathcal{M}_\Lambda$ is its Schur–Weyl multiplicity space [8, 9]. For perfectly indistinguishable photons, only the fully symmetric representation contributes, while partial indistinguishability allows contributions from other sectors. For two ports, the dominant contributions are concentrated in a window of $O(\sqrt{N})$ spin channels around j^*, which depends on the degree of indistinguishability [6]. For fixed M, the normalized

Young diagrams of the relevant sectors concentrate around the spectrum of the effective matrix $\rho_{\bar{n},\Gamma}$, in connection with the Keyl–Werner theorem and its extension to conditioning on fixed initial occupations [10]. However, this concentration does not determine the transition probabilities within each sector. For $M \geq 3$, weight spaces may exhibit internal multiplicity $m_\Lambda(n) > 1$, and the physical preparation conditioned on a sector of nonzero probability is described by the density matrix $\sigma_{\Lambda,n}(\Gamma)$, whose populations and coherences enter the transition probabilities. Determining their dependence on internal-state overlaps and the role of the different multiplicity directions is a central problem of this research.

The third correspondence connects the classical map with the semiclassical behavior of irreducible representations of $U(M)$. In the fully symmetric case, our previous work [7] used expansions based on virtual single-photon states to study multiphoton amplitudes through properties of single-photon transition amplitudes. For other representations, we have constructed pure states in the three-mode case by acting with the group on highest-weight states $|\Omega_\Lambda\rangle$ and projecting onto weight spaces $\mathcal{H}_{\Lambda,n}$. The compatibility of the frame with the initial occupations is equivalent to stationarity, at zero relative phases, of the action associated with the norm of the projected state. These constructions allow us to define sectorial classical maps $C_{\Lambda,V,U}(\phi)$ and establish exponential suppression bounds for transitions of projected states, provided that their initial norms are positive and not exponentially small. Building on these results, semiclassical approximations for representation matrix elements beyond the fully symmetric case will be formulated by identifying the contributions of stationary points, their prefactors, and the relative phases that determine their interference.

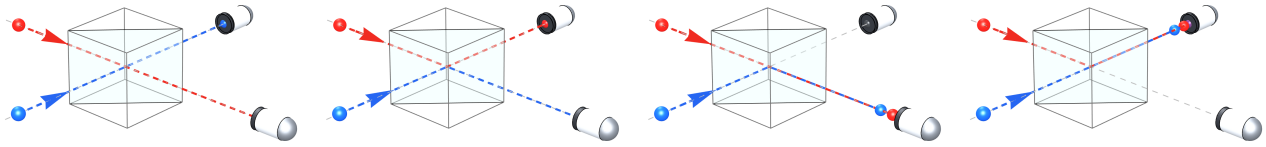
Together, these three correspondences define the central problem of this doctoral proposal: to establish, in the many-photon limit $N \to \infty$ with a fixed number of modes M, quantitative relations among the scattering probabilities of partially indistinguishable photons, the classical measure induced on the probability simplex by the map from relative phases to intensities, and the contributions of the irreducible sectors of $U(M)$. Although a correspondence has been established between the rescaled slowly varying component of the probabilities and the classical density at regular points, alongside an exact sector decomposition and concentration bounds for the statistical weights of the sectors under specific assumptions, the asymptotic behavior of the conditional transition probabilities and how their contributions relate to the full multiphoton distribution remain to be determined. To this end, we will determine how internal-state overlaps, described by the Gram matrix Γ, affect the populations and coherences of the physical density matrices $\sigma_{\Lambda,n}(\Gamma)$ in weight spaces with internal multiplicity. Furthermore, a construction based on multicoherent states will be formulated to parametrize their internal directions and establish the conditions under which they are related to the classical map. The semiclassical contributions associated with different preimages of the map and their relative phases will be calculated, and uniform Airy-type approximations near fold caustics and WKB-type approximations for classically forbidden regions will be formulated in configurations where they are applicable. Finally, semiclassical approximations for matrix elements of irreducible representations beyond the fully symmetric case will be developed, utilizing virtual states and constructions based on projected states to determine the contributions of stationary points, their prefactors, and relative phases.

1 Physical motivation and the many-photon regime

In multiphoton interference, different propagation alternatives may lead to the same detection outcome. If these alternatives are indistinguishable, their amplitudes add coherently, and the probability is the squared modulus of their sum. Thus, even in simple optical networks, interference can enhance or suppress outcomes that cannot be inferred from single-photon probabilities. Quantitatively determining this behavior as the number of photons increases, and establishing its correspondence with semiclassical descriptions, constitutes the central motivation of this research.

1.1 From two photons to interference in multiple modes

The Hong–Ou–Mandel effect shows how indistinguishability changes the detection probabilities of two photons [11]. Consider two perfectly indistinguishable photons entering different ports of a balanced beam splitter simultaneously. Fig. 2 shows the four alternatives involving transmission (T) and reflection (R), associated with their input ports.



(a) Both transmitted (TT); $(1,1)$.

(b) Both reflected (RR); $(1,1)$.

(c) Red T, blue R; $(0,2)$.

(d) Red R, blue T; $(2,0)$.

Figure 2: Propagation alternatives for two indistinguishable photons in a balanced beam splitter. The letters T and R denote transmission and reflection, respectively. The alternatives TT and RR lead to the output occupation $(1,1)$, and their amplitudes interfere destructively, while TR and RT lead to the occupations $(0,2)$ and $(2,0)$. The colors indicate the paths associated with the input ports and do not represent distinguishable internal states.

To detect one photon at each output, both photons can be transmitted (TT) or reflected (RR). The coincidence amplitude is the coherent sum of these two contributions. For the single-photon transformation of a balanced beam splitter, we choose

$$U_{\mathrm{BS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathcal{A}_{1,1} = U_{11}U_{22} + U_{12}U_{21} = 0. \tag{1.1}$$

The amplitudes TT and RR have equal magnitudes and opposite signs, so the coincidence amplitude vanishes through destructive interference. If the internal states are orthogonal, the probabilities of the two alternatives add, giving a coincidence probability of $1/2$. The bunching of indistinguishable photons results from interference, not from an attractive interaction.

Even with fixed M, increasing N can increase the number of alternatives leading to the same output occupation. Their interference depends on the relative phases and on the input and output occupations. For perfectly indistinguishable photons, transition amplitudes are expressed in terms of permanents of matrices constructed from U, with rows and columns repeated according to the occupations [12]. Their direct evaluation can become computationally costly as the number of photons N increases.

Permanents also play a central role in *boson sampling*, where the number of modes may grow with N [3]. In contrast, this proposal addresses the asymptotic limit $N \to \infty$ with a fixed number of modes M. Before formalizing this regime, the input preparation and detection model with partial distinguishability must be specified.

1.2 Preparation, partial distinguishability, and output probabilities

Consider a linear, passive, lossless interferometer with M ports, described at the single-photon level by a unitary matrix $U \in U(M)$. At the input, N photons are distributed according to the occupation $\boldsymbol{n} = (n_1, \ldots, n_M)$, with $\sum_i n_i = N$. The interferometer acts on the spatial modes, while the detectors record the output occupation $\boldsymbol{n}' = (n'_1, \ldots, n'_M)$, with $\sum_i n'_i = N$, without resolving internal degrees of freedom such as polarization, frequency, or temporal profile.

The analysis is restricted to the family of preparations considered in our previous work: the n_i photons entering port i share a pure, normalized internal state $|\chi_i\rangle$, while states associated with different ports may partially overlap. This assumption excludes arbitrary internal preparations, such as those with varying states for photons entering the same port or correlations not described by a common state for each port. Within this family, distinguishability between ports is characterized by the Gram matrix

$$\Gamma_{ij} = \langle \chi_j | \chi_i \rangle, \qquad \Gamma \succeq 0, \qquad \Gamma_{ii} = 1. \tag{1.2}$$

When all internal states coincide, $\Gamma_{ij} = 1$ and the photons are perfectly indistinguishable. At the opposite extreme, $\Gamma = I$ corresponds to orthogonal internal states for photons entering different ports, although photons entering the same port remain indistinguishable from one another. For partial overlaps, the complex entries of Γ determine which contributions can interfere coherently and how they combine in the detection probabilities [1, 2].

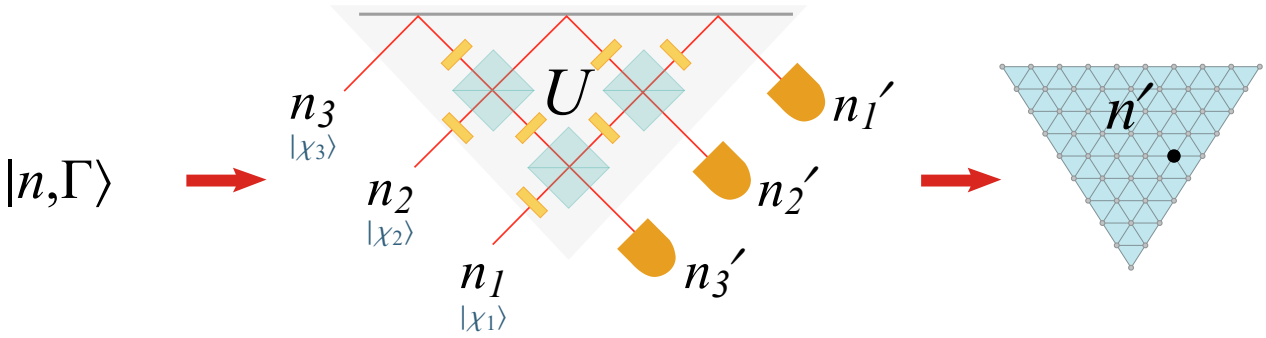


Figure 3: Schematic of the family of preparations considered in this work for a three-port interferometer. The n_i photons entering port i share an internal state $|\chi_i\rangle$, so partial distinguishability between ports is encoded in the overlaps $\Gamma_{ij} = \langle \chi_j | \chi_i \rangle$. The interferometer, described by the unitary matrix U, acts only on the spatial modes, while the detectors record the output occupations without resolving the internal states. Figure adapted from Fig. 1 of [7].

As illustrated in Fig. 3, for the same input occupation and interferometer U, changing the internal states $|\chi_i\rangle$ changes the overlaps in Γ and may change the output probabilities. The probability of detecting the occupation $\boldsymbol{n}'$ is denoted by $P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma)$. Although exact expressions are available, the presence of spatial and internal degrees of freedom makes it difficult to identify interference patterns as the number of photons increases [1, 2]. The objective is to establish the asymptotic output distribution beyond individual probabilities. To achieve this, the input occupations and the Gram matrix Γ will permit the formulation of an effective single-photon matrix which, together with the unitary U, defines the continuous classical map addressed in the next chapter.

1.3 The many-photon limit with a fixed number of modes

To quantify the macroscopic output distribution, the limit $N \to \infty$ is evaluated with M, U, and Γ fixed, and sequences of input and output occupations whose fractions converge to $\overline{\boldsymbol{n}}$ and $\overline{\boldsymbol{n}}'$, respectively:

$$N \longrightarrow \infty, \qquad M \text{ fixed}, \qquad \frac{\boldsymbol{n}}{N} \longrightarrow \overline{\boldsymbol{n}}, \qquad \frac{\boldsymbol{n}'}{N} \longrightarrow \overline{\boldsymbol{n}}'. \tag{1.3}$$

The normalized output occupations belong to the probability simplex $\Delta_{M-1} = \{\overline{\boldsymbol{n}}' : \overline{n}'_i \geq 0,\ \sum_i \overline{n}'_i = 1\}$. For $M = 3$, the simplex is a triangle whose points represent the fractions of photons detected at each port. This case allows us to study caustics, possible interior voids, and weight spaces with nontrivial internal multiplicity. For finite N, the possible occupations form a discrete lattice on the simplex that becomes finer as N increases. This enables the determination of regions of asymptotic concentration or strong suppression on a macroscopic scale, while retaining the structure of interference oscillations on smaller scales.

Although the number of modes remains fixed, this does not eliminate quantum interference or make the calculation of individual probabilities trivial. For two ports, the asymptotic correspondence between multiphoton interference with partial distinguishability and a classical description was established in [6]. Our subsequent work extended this framework to multimode interferometers for a specific family of preparations, demonstrating that the classical measure induced by the map from relative phases to intensities governs the macroscopic behavior of the output probabilities [7].

The auxiliary relative phases parametrize this map, and semiclassical approximation tools will permit the analytical calculation of the residual quantum oscillations over the continuum. The next chapter presents the formal construction of this classical map and establishes the conditions of validity for such asymptotic correspondence.

2 Semiclassical structure of multiphoton interference

This chapter presents the correspondence between multiphoton probabilities and the classical measure induced by the map from relative phases to intensities, established for two ports and extended to multimode interferometers in our previous work [6, 7]. In the limit $N \to \infty$ with fixed M, we present the established results and their conditions of validity, as well as open problems concerning caustics and the determination of suppression rates in classically forbidden regions.

2.1 Effective single-photon matrix and relative phases

For the preparations of Sec. 1.2, the normalized occupations $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ and the Gram matrix Γ define the effective single-photon matrix,

$$\rho_{\overline{\boldsymbol{n}},\Gamma} = [\overline{\boldsymbol{n}}]^{1/2}\,\Gamma\,[\overline{\boldsymbol{n}}]^{1/2}, \qquad [\overline{\boldsymbol{n}}] = \mathrm{diag}(\overline{n}_1, \ldots, \overline{n}_M). \tag{2.1}$$

Since $\Gamma \succeq 0$ and $\Gamma_{ii} = 1$, this matrix is positive semidefinite and satisfies

$$\mathrm{diag}\,\rho_{\overline{\boldsymbol{n}},\Gamma} = \overline{\boldsymbol{n}}, \qquad \mathrm{Tr}\,\rho_{\overline{\boldsymbol{n}},\Gamma} = 1. \tag{2.2}$$

The off-diagonal elements of $\rho_{\overline{\boldsymbol{n}},\Gamma}$ contain the internal-state overlaps weighted by the occupations. In the classical description, $\rho_{\overline{\boldsymbol{n}},\Gamma}$ is a normalized coherence matrix: its diagonal contains the relative input intensities, while its off-diagonal elements describe correlations between fields. It should not be confused with the reduced density matrix of the fixed-occupation multiphoton state.

In the construction based on virtual states, $\rho_{\overline{\boldsymbol{n}},\Gamma}$ is the spatial marginal of the auxiliary single-photon state at zero phases. The exact toroidal expansion of multiphoton states in tensor powers of these states leads to a double-integral representation of the probabilities (Propositions 1 and 2 of Ref. [7]). The phase matrix is defined as

$$t_{\boldsymbol{\phi}} = \mathrm{diag}\left(e^{\mathrm{i}\phi_1}, \ldots, e^{\mathrm{i}\phi_M}\right). \tag{2.3}$$

In the quantum expansion, these phases are auxiliary parameters and do not correspond to phases experimentally applied to the fixed-occupation state; in the classical description, they represent relative input phases. Removing the global phase leaves $M-1$ independent phases forming $\mathbb{T}^{M-1}$. Varying these phases with U and $\rho_{\overline{\boldsymbol{n}},\Gamma}$ fixed defines the classical map.

2.2 Classical map, allowed region, and induced measure

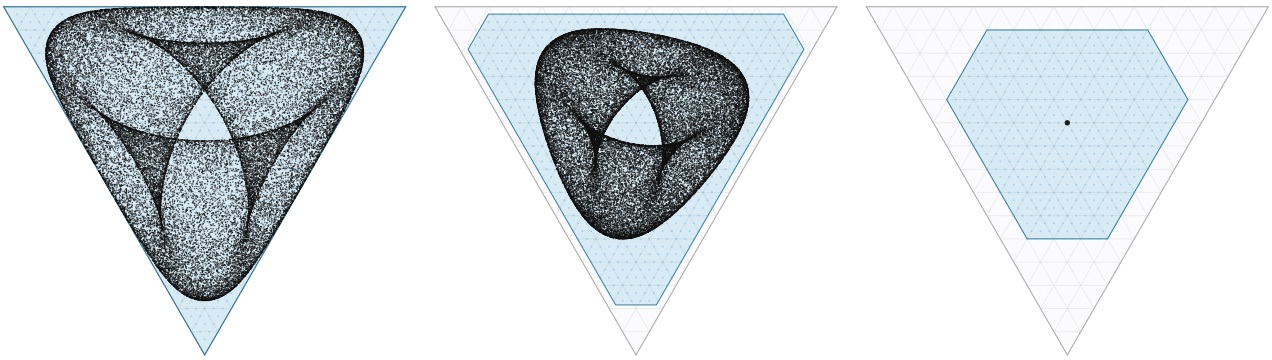
The propagation of the effective matrix through U, modulated by the phases $\boldsymbol{\phi}$, defines the classical map

$$C_{\rho,U}(\boldsymbol{\phi}) = \mathrm{diag}\Big(U t_{\boldsymbol{\phi}} \rho_{\overline{\boldsymbol{n}},\Gamma} t_{\boldsymbol{\phi}}^{\dagger} U^{\dagger}\Big), \qquad C_{\rho,U} : \mathbb{T}^{M-1} \longrightarrow \Delta_{M-1}, \tag{2.4}$$

where Δ_{M-1} is the simplex of normalized intensities. The image of the map is the classically allowed region,

$$\mathcal{R}_{\rho,U} = C_{\rho,U}(\mathbb{T}^{M-1}) \subseteq \Delta_{M-1}. \tag{2.5}$$

For $M = 3$, Fig. 4 compares the images of the map on the triangular simplex for three degrees of indistinguishability, with the input occupation and interferometer fixed. By the Schur–Horn theorem, these images are contained in the corresponding spectral permutahedra, shown in blue, but need not coincide with them.



(a) Perfect indistinguishability. (b) Partial distinguishability. (c) Orthogonal internal states between ports.

Figure 4: Images of the classical map for $N = 45$ photons, with input occupation $\boldsymbol{n} = (30, 12, 3)$ and $U = F_3$, for three degrees of indistinguishability: (a) perfectly indistinguishable photons, $\Gamma_{ij} = 1$; (b) partial indistinguishability, $\Gamma = \Gamma_{\mathrm{par}}$, with $(\Gamma_{\mathrm{par}})_{12} = (\Gamma_{\mathrm{par}})_{13} = 1/\sqrt{2}$ and $(\Gamma_{\mathrm{par}})_{23} = (3 - \mathrm{i}\sqrt{3})/8$, together with unit diagonal entries and conjugate off-diagonal entries; and (c) orthogonal internal states between ports, $\Gamma = I_3$. The black points are images of relative phases sampled on the torus, while the blue regions represent the corresponding spectral permutahedra.

The induced classical measure is obtained by pushing forward the normalized uniform measure on the torus to the simplex through $C_{\rho,U}$. At regular points $\overline{\boldsymbol{n}}'$, with discrete preimages and a nonzero reduced Jacobian, its density with respect to the independent simplex coordinates is

$$f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma) = \frac{\mathbf{1}_{\mathcal{R}_{\rho,U}}(\overline{\boldsymbol{n}}')}{(2\pi)^{M-1}} \sum_{\boldsymbol{\phi}_\alpha :\, C_{\rho,U}(\boldsymbol{\phi}_\alpha) = \overline{\boldsymbol{n}}'} \left| \det\left(\frac{\partial C_{\rho,U}}{\partial \boldsymbol{\phi}} \right)_{\boldsymbol{\phi}_\alpha} \right|^{-1}. \tag{2.6}$$

The determinant is the reduced Jacobian in independent coordinates (for example, $\phi_1 = 0$ and components $C_2, \ldots, C_M$); each preimage contributes the inverse of its absolute value. Caustics may arise where this Jacobian vanishes, at critical points where the regular formula fails and a different asymptotic treatment is needed. For $M \geq 3$, interior caustics and voids (interior regions without preimages) may occur. If $\Gamma = I_3$ and $U = F_3$, the image is the single point $(1/3, 1/3, 1/3)$, and the measure is concentrated at that point. The map delimits the forbidden region but does not, by itself, determine the quantum probabilities there or their behavior near caustics; the following theorems specify the scope of this correspondence [7].

2.3 Established asymptotic results

Three theorems of Ref. [7] make this correspondence precise.

Concentration in the classical region. Theorem 3 establishes the bound

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) \leq (N+1)^M \exp\left[-N D(\overline{\boldsymbol{n}}' \| \mathcal{R}_{\rho,U})\right], \tag{2.7}$$

where $D(\overline{\boldsymbol{n}}' \| \mathcal{R}_{\rho,U}) = \min_{\boldsymbol{\nu} \in \mathcal{R}_{\rho,U}} D(\overline{\boldsymbol{n}}' \| \boldsymbol{\nu})$ and $D(\boldsymbol{x} \| \boldsymbol{y}) = \sum_i x_i \ln(x_i/y_i)$ is the relative entropy. At a positive distance from the allowed region, the bound implies exponential suppression as $N \to \infty$, but it determines neither the probabilities inside the region nor the exact suppression rates outside it. Its mathematical form is analogous to large-deviation bounds [13, 14].

Limit of the macroscopic envelope. P_{slow} collects the diagonal contributions of the double integral. At regular points, Theorem 4 shows that its rescaled density, $f_{\mathrm{slow}} = N^{M-1} P_{\mathrm{slow}}$, converges to the classical density:

$$\lim_{N\to\infty} f_{\mathrm{slow}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma) = f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma). \tag{2.8}$$

This limit does not, by itself, imply that quantum oscillations disappear.

Suppression of rapid interference oscillations. For generic interferometers and full-rank $\rho_{\overline{\boldsymbol{n}},\Gamma}$, Theorem 5 establishes the asymptotic suppression of the off-diagonal interference contributions, so that

$$\lim_{N\to\infty} N^{M-1} P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma), \tag{2.9}$$

both at regular points of the allowed region and in the forbidden region. This limit now concerns the rescaled physical probabilities.

If the initial occupation fractions are strictly positive, $\rho_{\overline{\boldsymbol{n}},\Gamma}$ and Γ have the same rank. Rank deficiency of Γ is necessary, but not sufficient, for rapid interference oscillations to persist asymptotically; perfect indistinguishability ($\operatorname{rank}\Gamma = 1$) provides an example. These theorems do not provide uniform approximations near caustics or a general characterization of persistent interference.

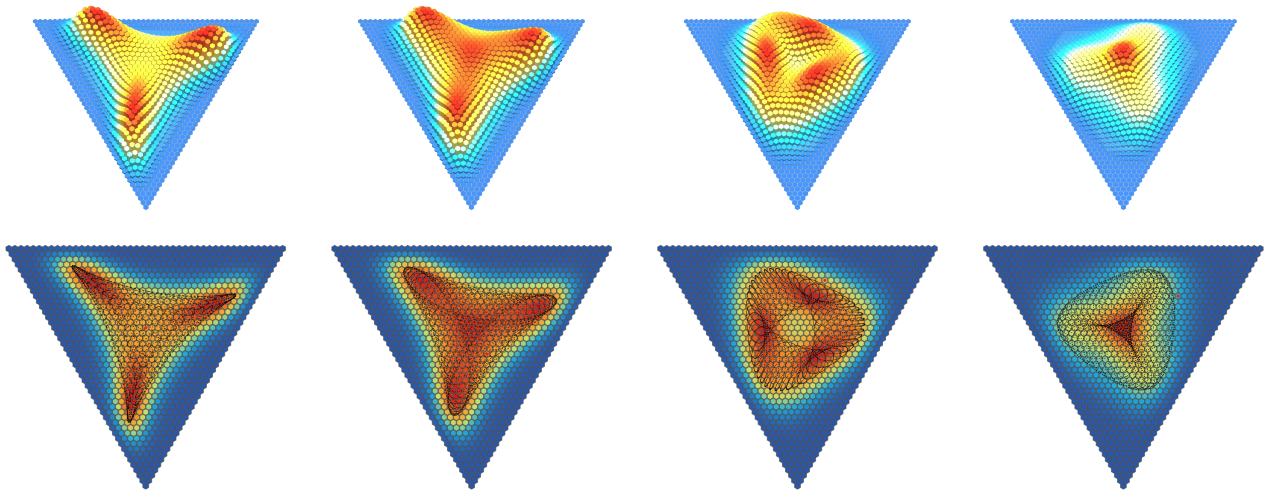
2.4 Comparison with quantum probabilities and scope of the correspondence

To compare multiphoton probabilities with the images of the classical map, four preparations are considered with $N = 45$, $U = F_3$, and a common effective spectrum $\boldsymbol{\lambda} = (11, 3, 1)/15$ (Table 1).

Table 1: Input occupations of the four preparations considered in Fig. 5, for $N = 45$ photons. The occupations $\boldsymbol{n}$ and their normalized forms $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ are shown.

Panel	$\boldsymbol{n}$	$\overline{\boldsymbol{n}}$
(a)	$(15, 15, 15)$	$(1, 1, 1)/3$
(b)	$(15, 10, 20)$	$(3, 2, 4)/9$
(c)	$(24, 15, 6)$	$(8, 5, 2)/15$
(d)	$(27, 9, 9)$	$(3, 1, 1)/5$

By the Schur–Horn theorem, $\overline{\boldsymbol{n}}^{(j)} \prec \boldsymbol{\lambda}$ guarantees the existence of effective matrices with that spectrum and diagonal, without uniquely determining their internal-state overlaps. Fig. 5 compares the output probabilities with sampled images of the map on the simplex.



(a) Balanced $\boldsymbol{n} = (15, 15, 15)$.

(b) Asymmetric $\boldsymbol{n} = (15, 10, 20)$.

(c) Intermediate $\boldsymbol{n} = (24, 15, 6)$.

(d) More pronounced $\boldsymbol{n} = (27, 9, 9)$.

Figure 5: Comparison between multiphoton detection probabilities and the classical map for four preparations of $N = 45$ photons, with $U = F_3$ and a common effective spectrum $\boldsymbol{\lambda} = (11, 3, 1)/15$. The upper panels show the output probabilities using heights and colors, while the lower panels show the same probabilities on the simplex of normalized occupations, together with the black points obtained by sampling the classical map. The input occupations are indicated below each column.

For $\boldsymbol{\lambda} = \operatorname{spec} \rho_{\overline{\boldsymbol{n}},\Gamma}$, the Schur–Horn theorem implies

$$\mathcal{R}_{\rho,U} \subseteq \Pi(\boldsymbol{\lambda}), \qquad \Pi(\boldsymbol{\lambda}) = \operatorname{conv}\{\pi\boldsymbol{\lambda} : \pi \in S_M\}, \tag{2.10}$$

The spectral permutahedron $\Pi(\boldsymbol{\lambda})$ does not necessarily coincide with the image of the map [7, 15]. With the same spectrum and interferometer, the four preparations give different images and quantum distributions whose macroscopic patterns reflect these differences. Thus, the spectrum alone determines neither the image of the map nor the output probabilities.

The comparison for $N = 45$ illustrates, but does not prove, the asymptotic limits of Sec. 2.3. At regular points, the induced classical measure describes the macroscopic envelope, but does not determine the individual contributions of the irreducible sectors of $U(M)$ or semiclassical approximations for matrix elements beyond the fully symmetric representation. These connections provide the basis for the following chapters.

3 Multiphoton interference and representation theory

Chapter 2 related multiphoton probabilities to the measure induced by the classical map [7]. Starting from this macroscopic description, we present the exact decomposition into irreducible sectors of $U(M)$, whose internal structure allows us to determine conditional probabilities that the classical map itself does not specify.

3.1 Input configuration, interference, and irreducible sectors

Fig. 6 relates the input preparation to the classical map, multiphoton probabilities, and their irreducible sectors.

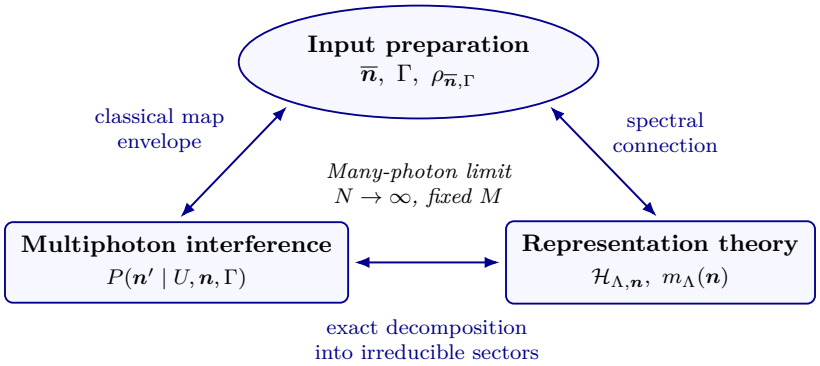


Figure 6: Relations among the input preparation, multiphoton interference, and representation theory. The connections correspond to the classical map, the spectral relation, and the exact decomposition into irreducible sectors, whose asymptotic behavior is a central problem of this research.

The image of the classical map and its induced measure describe the allowed region and the macroscopic envelope under the assumptions of Chapter 2 [7]. The spectrum of the effective matrix is related to the concentration of sector weights, even for a fixed initial occupation (Appendix A), but does not determine the internal distributions. The exact decomposition leaves open the relation between the probabilities of each sector, weight-space multiplicities, and the semiclassical behavior of representation functions.

3.2 Irreducible sectors, spectrum, and weight diagrams

With partial distinguishability, the reduced spatial description includes sectors that are not fully symmetric, although the full state retains its bosonic symmetry. Schur–Weyl duality organizes them as

$$(\mathbb{C}^M)^{\otimes N} \simeq \bigoplus_{\substack{\Lambda \vdash N \\ \ell(\Lambda) \le M}} \mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda, \tag{3.1}$$

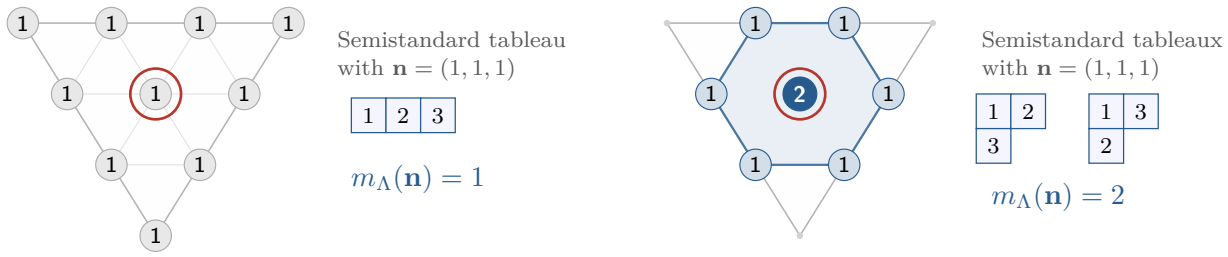
Each partition $\Lambda = (\Lambda_1, \ldots, \Lambda_M)$ of N labels an irreducible representation of $U(M)$ acting on $\mathcal{H}_\Lambda$, with multiplicity space $\mathcal{M}_\Lambda$ [8, 9]. For $M = 3$, its $SU(3)$ Dynkin labels are $(p, q) = (\Lambda_1 - \Lambda_2, \Lambda_2 - \Lambda_3)$. The sector weights are related to $\boldsymbol{\lambda} = \operatorname{spec} \rho_{\overline{\boldsymbol{n}},\Gamma}$. For product states $\rho^{\otimes N}$, Keyl and Werner proved the concentration of $\overline{\Lambda} = \Lambda/N$ around $\operatorname{spec} \rho$ [16]. When $\rho_{\overline{\boldsymbol{n}},\Gamma}^{\otimes N}$ is conditioned on the fixed occupation $\boldsymbol{n}$, whose probability is at least $(N+1)^{-M}$, the exponential bounds remain valid up to a polynomial factor (Appendix A). This concentration concerns the normalized irreducible labels, but does not determine the distribution within the weight spaces or the transition probabilities conditioned on each sector.

Each representation decomposes into weight spaces associated with the occupations:

$$\mathcal{H}_\Lambda = \bigoplus_{\boldsymbol{n}} \mathcal{H}_{\Lambda,\boldsymbol{n}}, \qquad m_\Lambda(\boldsymbol{n}) = \dim \mathcal{H}_{\Lambda,\boldsymbol{n}}. \tag{3.2}$$

The weight multiplicity $m_\Lambda(\boldsymbol{n})$ counts the internal directions associated with an occupation, distinct from the Schur–Weyl multiplicity $\dim \mathcal{M}_\Lambda$. The detectors do not resolve these directions, although the evolution may couple them.

The number of semistandard Young tableaux of shape Λ and content $\boldsymbol{n}$ is $m_\Lambda(\boldsymbol{n})$. For $M = N = 3$, Fig. 7 shows ten weights of multiplicity one for $\Lambda = (3,0,0)$ and seven for $\Lambda = (2,1,0)$, with multiplicity two at $\boldsymbol{n} = (1,1,1)$. The convex hull of the normalized weights is the permutahedron $\Pi(\overline{\Lambda})$, which generally differs from $\Pi(\boldsymbol{\lambda})$ and the classical image $\mathcal{R}_{\rho,U}$. These internal multiplicities occur for $U(3)$, but not in irreducible representations of $U(2)$.



(a) Symmetric: $\Lambda = (3,0,0)$. (b) Adjoint: $\Lambda = (2,1,0)$.

Figure 7: Weight diagrams and semistandard Young tableaux of two representations of $U(3)$ for $N = 3$: (a) fully symmetric representation, $\Lambda = (3,0,0)$; (b) representation $\Lambda = (2,1,0)$, corresponding to the adjoint of $SU(3)$. The numbers indicate the weight multiplicities. The marked weight, $\mathbf{n} = (1,1,1)$, has multiplicity one in (a) and two in (b), as shown by the corresponding tableaux.

3.3 From the symmetric case to multiplicity spaces

For perfectly indistinguishable photons, each occupation corresponds to a unique Fock state in $\mathcal{H}_{(N,0,\ldots,0)} = \mathrm{Sym}^N(\mathbb{C}^M)$; the amplitudes can be expressed in terms of permanents or representation matrix elements [12]. Semiclassical approximations are available for Wigner functions of $SU(2)$ and $SU(3)$ [17, 18]. In our previous work, known bosonic formulas were recovered using virtual single-photon states [7], motivating their extension to other representations.

With partial distinguishability, transition probabilities also depend on the preparation and the internal directions of each weight space.

The preparation is described by the normalized physical density matrix

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \frac{P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}}}{\mathrm{Tr}\left[P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}}\right]}, \tag{3.3}$$

Here $P_{\Lambda,\boldsymbol{n}}$ projects onto $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, and $D^\Lambda(\Gamma)$ is the polynomial extension of the representation to Γ, which is not necessarily unitary. The density matrix is defined when $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$ and is positive semidefinite with unit trace (Appendix A). In a weight-space basis, its diagonal and off-diagonal entries represent populations and coherences, respectively.

If Γ is positive definite, $\operatorname{rank} \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = m_\Lambda(\boldsymbol{n})$ (Appendix G); for $m_\Lambda(\boldsymbol{n}) > 1$, the physical density matrix is mixed and cannot be identified with a projected pure state. Parametrizing the internal directions using suitable preparations, compatible frames, and projected states is a central problem motivating the next chapter.

3.4 Physical probability as a sum of sector contributions

The exact decomposition of the physical probability is (Appendices A and G):

$$\begin{aligned} P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) &= \sum_\Lambda P(\Lambda \mid \boldsymbol{n}, \Gamma)\, P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma), \\ P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma) &= \mathrm{Tr}\Big[P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) D^\Lambda(U)^\dagger\Big]. \end{aligned} \tag{3.4}$$

The first factor corresponds to the sector weight; the second, defined when $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$, is the conditional probability. Only sectors with nonzero weight contribute to the sum; the trace accounts for internal directions not resolved by the detectors. The identity is exact for finite N.

The exact decomposition does not determine the asymptotic behavior of the conditional probabilities or how their contributions reconstruct the macroscopic distribution described by the classical map of Chapter 2. When all normalized initial occupations are positive, $\rho_{\overline{\boldsymbol{n}},\Gamma}$ allows the reconstruction of Γ; however, characterizing transitions among the internal directions of weight spaces in the semiclassical limit remains an open problem. The next chapter introduces compatible frames, projected pure states, and sectorial classical maps as analytical tools to address this problem, without identifying these pure states with the physical sector density matrix.

4 Progress toward a sectorial semiclassical description

Chapter 3 identified the asymptotic behavior of transitions between weight spaces with internal multiplicity $m_\Lambda(\boldsymbol{n}) > 1$ as an open problem. For the three-mode case, we present compatible frames, projected pure states, and sectorial maps, distinguishing exact results, numerical evidence, and open asymptotic problems.

4.1 Compatible frames and spectrum–diagonal geometry

Let $\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3)$, $\sum_i \Lambda_i = N$, be a partition labeling an irreducible representation of $U(3)$, with $\overline{\Lambda} = \Lambda/N$. For $V \in U(3)$, the sectorial matrix is defined as

$$\rho_\Lambda(V) = V[\overline{\Lambda}]V^\dagger, \qquad Q(V)_{ij} = |V_{ij}|^2, \qquad \operatorname{diag}\rho_\Lambda(V) = Q(V)\overline{\Lambda}, \tag{4.1}$$

where $[\overline{\Lambda}]$ is diagonal and $Q(V)$ is unistochastic, and therefore bistochastic.

The frame V is compatible with $\overline{\boldsymbol{n}}$ if

$$Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}. \tag{4.2}$$

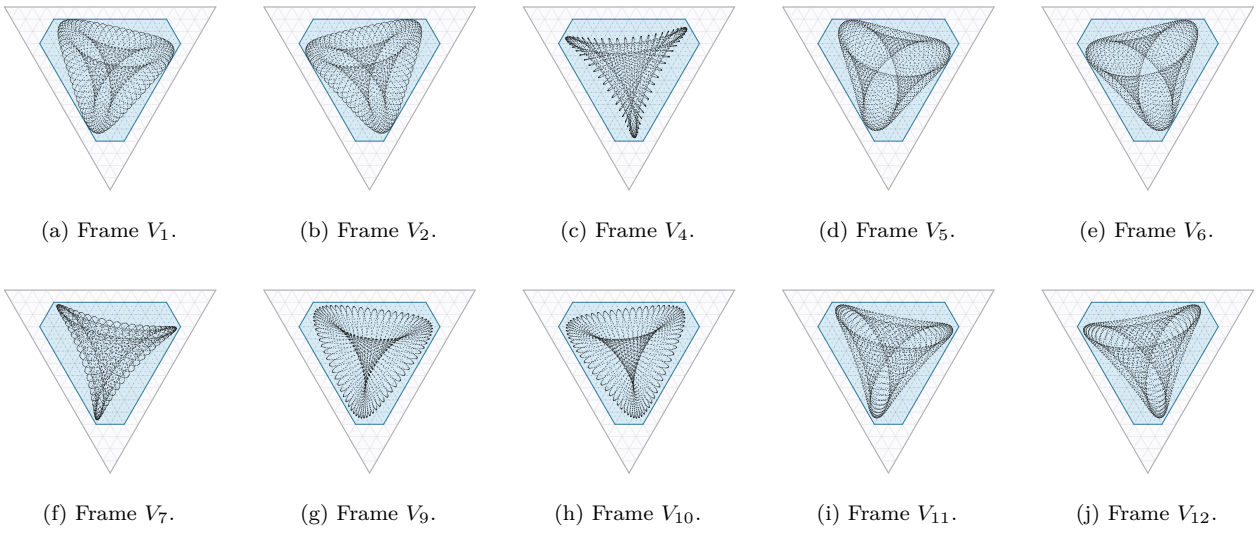
The Schur–Horn theorem guarantees the existence of compatible frames if and only if $\overline{\boldsymbol{n}} \prec \overline{\Lambda}$, without determining V or the off-diagonal elements of $\rho_\Lambda(V)$ [15, 19]. For three modes, the bistochastic matrices Q satisfying $Q\overline{\Lambda} = \overline{\boldsymbol{n}}$ are characterized, and unistochasticity is imposed to reconstruct the frames [20]; Heron's criterion for triangle closure is detailed in Appendix B [21, 22].

The sectorial matrix $\rho_\Lambda(V)$ does not necessarily coincide with the physical effective matrix $\rho_{\overline{\boldsymbol{n}},\Gamma}$. If $\overline{n}_i > 0$ for all i, each compatible frame defines an admissible Gram matrix,

$$\Gamma(V) = [\overline{\boldsymbol{n}}]^{-1/2}\rho_\Lambda(V)[\overline{\boldsymbol{n}}]^{-1/2}. \tag{4.3}$$

Different compatible frames may produce different matrices $\Gamma(V)$, even with the same spectrum and initial occupation. Agreement with the physical matrix requires $\overline{\Lambda} = \operatorname{spec}\rho_{\overline{\boldsymbol{n}},\Gamma}$ and that V diagonalizes it. In this case, the classical maps coincide, but the transition probabilities of the projected states and of the physical sectorial density matrix do not necessarily coincide.

Fig. 8 compares ten compatible frames with fixed Λ, $\overline{\boldsymbol{n}}$, and U. Although they share the same spectrum and diagonal, their images under the sectorial map differ because of the dependence on the frame V and the associated matrix $\Gamma(V)$. The points represent normalized classical intensities, not detection probabilities; the examples do not constitute an exhaustive classification.



(a) Frame V_1. (b) Frame V_2. (c) Frame V_4. (d) Frame V_5. (e) Frame V_6.

(f) Frame V_7. (g) Frame V_9. (h) Frame V_{10}. (i) Frame V_{11}. (j) Frame V_{12}.

Figure 8: Images of the sectorial classical map for ten compatible frames, with $N = 45$, $\Lambda = (33, 9, 3)$, $\overline{\boldsymbol{n}} = (1/3, 2/9, 4/9)$, and $U = F_3$. In each panel, the black points represent samples obtained by applying the map to relative phases on the torus, while the blue region corresponds to the common spectral permutahedron $\Pi(\overline{\Lambda})$. Although the spectrum, initial occupation, and interferometer remain fixed, the geometry of the image depends on the frame V.

4.2 Projected states and multiplicity directions in $SU(3)$

To explore other representations, the highest-weight orbit states $D^{\Lambda}(V)|\Omega_{\Lambda}\rangle$ are projected onto the occupation $\boldsymbol{n}$ [23]. This projection admits an exact toroidal representation (Appendix C), although its general semiclassical approximation remains an open problem. The normalized state is

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V)|\Omega_{\Lambda}\rangle}{\sqrt{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)}}, \qquad \mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \left\| P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V)|\Omega_{\Lambda}\rangle \right\|^2, \tag{4.4}$$

provided that $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) > 0$. For $SU(3)$, Gelfand–Tsetlin and isospin bases are used (Appendices C–F) [24, 25, 26].

In the adjoint representation of $SU(3)$, with $\Lambda = (2, 1, 0)$ and $\boldsymbol{n} = (1, 1, 1)$, two frames with the same $Q(V)$ are constructed whose projected states are orthogonal (Appendix F). This exact result demonstrates the importance of the phases of V, without implying differences between their isospin populations.

The linear independence of projected states is examined through their Gram matrices. For $N = 45$, $\Lambda = (33, 9, 3)$, families of states are found numerically to span the seven-dimensional weight spaces corresponding to $\boldsymbol{n} = (15, 15, 15)$, $(27, 9, 9)$, and $(15, 10, 20)$ (Appendix F). The rank criterion is exact, but the observed spanning is based on numerical evidence and does not guarantee that every direction can be obtained from a single frame.

The projected states are pure and are not, in general, equivalent to the physical sectorial density matrix $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. Their transition probabilities are given by

$$P_{\mathrm{marco}}(\boldsymbol{n}') = \left\| P_{\Lambda,\boldsymbol{n}'} D^{\Lambda}(U)|\boldsymbol{n}, V; \Lambda\rangle \right\|^2, \tag{4.5}$$

The conditional physical probabilities follow Eq. (3.4); for $N = 45$, the two descriptions are compared numerically and the isospin distributions are analyzed. The projected states serve as analytical probes of internal multiplicity. A continuous parametrization of these directions through multicoherent states of $SU(M)$ is proposed as a way to establish their semiclassical limit, without identifying this limit directly with the physical sector density matrix.

4.3 Sectorial map and scope of the asymptotic results

In the toroidal representation, the compatibility condition $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ is equivalent to the stationarity of the action for the projected norm at the origin of the relative phases. If the reduced Hessian is positive definite and the required regularity conditions hold, a local Gaussian approximation is obtained. Extending it to the full integral requires controlling the remaining contributions (Appendix H). The macroscopic analysis of the transitions leads to the classical map associated with the sector (Appendix I):

$$C_{\Lambda,V,U}(\boldsymbol{\phi}) = Q(Ut_{\boldsymbol{\phi}}V)\overline{\Lambda} = \operatorname{diag}\left[Ut_{\boldsymbol{\phi}}\rho_{\Lambda}(V)t_{\boldsymbol{\phi}}^{\dagger}U^{\dagger}\right], \qquad \mathcal{R}_{\Lambda,V,U} = C_{\Lambda,V,U}(\mathbb{T}^{M-1}). \tag{4.6}$$

By the Schur–Horn theorem, $\mathcal{R}_{\Lambda,V,U} \subseteq \Pi(\overline{\Lambda})$, although the image does not necessarily coincide with the permutahedron. The map identifies the region where the exponential penalty for transitions from projected pure states may vanish, but does not by itself determine their probabilities.

For normalized final occupations at a uniformly positive distance from $\mathcal{R}_{\Lambda,V,U}$, Appendix I establishes an exponential suppression bound, provided that the initial projected norms are positive and not exponentially small. Unlike the Gaussian approximations, this bound does not require regularity of the map. The determination of the prefactors, the interference between stationary points, and a general semiclassical approximation for the conditional probabilities of the physical density matrix $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ remain open problems.

Supplementary material and reproducibility. The mathematical appendices accompanying this proposal and the Wolfram Language code associated with the numerical calculations presented in the document are available in the GitHub repository: `https://github.com/orlandomelovargas/Multiphoton-Interference-Candidacy`.

5 Research problem and objectives

The preceding chapters presented the semiclassical correspondence between multiphoton probabilities and the measure induced by the classical map, the exact decomposition of these probabilities into irreducible sectors, and the constructions based on projected states and sectorial maps. These results provide a starting point for investigating how sector contributions relate to the multiphoton distribution and how classical maps can guide the development of semiclassical approximations for representation matrix elements.

5.1 Statement of the scientific problem

In the many-photon limit, we showed that probabilities are exponentially suppressed at a positive distance from the image of the classical map from relative phases to intensities, and that their rescaled slowly varying component converges to the density of the induced classical measure at regular points [7]. Under additional conditions on the interferometer and the rank of the effective matrix, this convergence also holds for the rescaled physical probabilities. Open problems include determining the exact asymptotic rate functions, describing the behavior near caustics and across the transition between classically allowed and forbidden regions, and characterizing the interference that may persist when the effective matrix is not of full rank.

Schur–Weyl duality provides an exact decomposition of the physical probabilities into irreducible sectors (Eq. (3.4)). The normalized Young diagrams labeling these sectors concentrate around the spectrum of the effective matrix, in connection with the Keyl–Werner theorem and the result for fixed initial occupations proved in Appendix A [16]. This concentration does not determine the probabilities conditioned on each sector or their contributions to the total distribution. It remains to characterize how internal-state overlaps affect the populations and coherences of the physical density matrices in weight spaces with multiplicity, and to explore their internal directions and role in transitions.

Virtual single-photon states enable the semiclassical study of amplitudes in the fully symmetric

representation [7]. For other representations, we have constructed projected pure states and sectorial classical maps (Eq. (4.6)), obtaining exponential suppression bounds for their transitions when the initial norms are positive and not exponentially small. The stationary contributions, their prefactors and interference remain to be determined, as does the relation between these constructions and the physical sector probabilities, without identifying projected pure states with physical density matrices.

The central problem is thus to relate, in the many-photon limit, the induced classical measure, multiphoton probabilities with partial indistinguishability, and the contributions of the irreducible sectors of $U(M)$, establishing the conditions under which semiclassical approximations can be obtained.

5.2 General objective

To establish quantitative correspondences among the scattering probabilities of partially indistinguishable photons in the many-photon limit $N \to \infty$ with a fixed number of modes M, the semiclassical behavior of irreducible representations of $U(M)$ in the limit of large Young diagrams, and classical multiport interferometry with partially coherent light, described by the classical map and its induced measure on the probability simplex.

5.3 Specific objectives

The specific objectives are organized around the three correspondences underlying this research.

1. Classical interferometry and multiphoton interference

To determine how the preimages and singularities of the classical map affect the asymptotic behavior of multiphoton probabilities, beginning with three-mode interferometers.

1.1. To identify and classify the singularities of the classical map in selected configurations through its preimages and Jacobian, and quantify their effects on the probabilities.

1.2. To determine bounds and decay rates for the probabilities in classically forbidden regions, establishing their relation to bounds based on relative entropy.

1.3. To formulate and test uniform Airy-type approximations near fold caustics and WKB connection formulas between allowed and forbidden regions, under appropriate conditions.

2. Multiphoton interference and representation theory

To quantify how partial indistinguishability determines the contributions of the irreducible sectors of $U(M)$ and their weight spaces in the limit of large Young diagrams, beginning with $U(3)$.

2.1. To characterize the distributions of statistical sector weights and quantify their concentration using Keyl–Werner bounds conditioned on fixed initial occupations.

2.2. To determine how the populations and coherences of the physical density matrices $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ enter the conditional probabilities and how these contribute to the full multiphoton distribution.

2.3. To formulate and evaluate constructions based on multicoherent states, compatible frames, and projected states to parametrize the internal directions of weight spaces and establish their relation to physical sector density matrices.

3. Classical interferometry and representation theory

To construct semiclassical approximations for matrix elements of irreducible representations of $U(M)$ beyond the fully symmetric sector, using virtual states, projections onto weight spaces, and classical maps associated with sectors.

3.1. To formulate suitable integral representations for matrix elements of representations beyond the fully symmetric case, identifying their stationarity conditions and their relation to sectorial classical maps.

3.2. To determine, for selected families of $U(3)$ representations, the contributions of stationary points, their prefactors and relative phases, and compare the asymptotic formulas with exact and numerical results.

5.4 Scope and expected contributions

The research will initially focus on three-mode interferometers and representations of $U(3)$, where weight spaces with internal multiplicity $m_\Lambda(\boldsymbol{n}) > 1$ occur and the classical map may have interior caustics. Using Gelfand–Tsetlin bases, the isospin structures of $SU(3)$, and the exact and numerical results already obtained, quantitative correspondences will be investigated in specific configurations, within the general framework of the limit $N \to \infty$ with fixed M.

For the first correspondence, the singularities of the classical map and their effects on multiphoton probabilities will be characterized, determining bounds and rates of exponential suppression, and formulating Airy- and WKB-type approximations in configurations where they apply. For the second, the contributions of irreducible sectors to the multiphoton distribution will be quantified, considering the concentration of their statistical weights, the populations and coherences of the physical density matrices $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$, and the use of multicoherent constructions to describe internal multiplicity directions.

Finally, for the third correspondence, semiclassical approximations for matrix elements of irreducible representations beyond the fully symmetric case will be constructed, identifying stationary contributions, prefactors, and relative phases through virtual and projected states. The analytical results will be compared with exact and numerical calculations, distinguishing physical sector probabilities from transitions of projected pure states. A general treatment of more highly degenerate singularities, complex stationary points, and the systematic extension to an arbitrary number of modes will remain possible directions for future work.

5.5 Methodology

The methodology will combine representation theory, semiclassical methods, and symbolic and numerical calculations, building on the exact results already developed. The research will initially focus on three-mode interferometers, considering sequences of occupations $\boldsymbol{n}$ and representations Λ such that $\boldsymbol{n}/N \to \overline{\boldsymbol{n}}$ and $\Lambda/N \to \overline{\Lambda}$ as $N \to \infty$, with fixed M.

For the first correspondence, the classical map $C_{\rho,U}$ will be analyzed through its preimages and Jacobians, identifying singularities, caustics, and voids. The induced classical measure will be compared with multiphoton probabilities for increasing values of N, examining contributions from different preimages and their relative phases where appropriate. Building on relative-entropy bounds, the rates of exponential suppression will be investigated and Airy- and WKB-type approximations will be formulated in configurations where the singularities and transitions between allowed and forbidden regions justify their use.

For the second correspondence, the exact decomposition of Eq. (3.4) and the concentration bounds associated with the Keyl–Werner theorem will be used. The statistical sector weights will be calculated and the conditional physical probabilities will be analyzed using $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ and its evolution under $D^\Lambda(U)$, examining how their populations and coherences contribute to the total distribution. For $SU(3)$, the isospin distributions and their moments will be determined, distinguishing the physical populations from the coefficients $c_j(V)$ of projected pure states. Through inner products and Gram matrices of these states, the generation of internal multiplicity directions and their possible description through multicoherent constructions will be evaluated.

Finally, for the third correspondence, we will implement constructions based on exterior powers, highest-weight orbits, and toroidal projections (Appendices C, H, and I), using the methods based on virtual single-photon states as a reference. Using stationarity conditions, reduced Hessians, and sectorial classical maps, we will develop semiclassical approximations for matrix elements of representations beyond the fully symmetric case in selected configurations. We will examine the contributions of stationary points, their prefactors and relative phases, together with the conditions of validity and

exponential suppression bounds. The results will be compared with exact and numerical calculations, distinguishing transitions of projected pure states from physical sector probabilities.

5.6 Ethical considerations

The research is theoretical and computational and therefore does not involve experiments with humans or animals or the processing of personal data. The main ethical considerations concern academic integrity and the reproducibility of the results. Proper attribution of previous work and documentation of the analytical and numerical procedures used will be strictly ensured. Furthermore, proved results, numerical evidence, and hypotheses yet to be verified will be explicitly distinguished, stating the conditions of validity and limitations of the approximations developed.

5.7 Timeline

The work plan covers doctoral semesters 7–10 and is organized around the three correspondences underlying the specific objectives. Building on the results already obtained, the analysis of the classical map, the characterization of sector contributions, and the development of semiclassical approximations for representation matrix elements will be carried out.

Table 2: Tentative timeline.

Period	**Planned activities and results**
Semester 7	Characterization of singularities, caustics, and voids of the classical map in three modes using preimages and Jacobians. Calculation of bounds and rates of exponential suppression, and evaluation of Airy/WKB approximations in appropriate configurations, with comparisons against exact probabilities.
Semester 8	Calculation of statistical sector weights and their concentration. Analysis of conditional physical probabilities, populations, and coherences of $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$, and exploration of multiplicity directions using compatible frames, projected states, and multicoherent constructions.
Semester 9	Development of semiclassical approximations for matrix elements of $U(3)$ using virtual states and toroidal projections. Determination of stationary contributions, prefactors, and relative phases. Quantification of their relation to sector probabilities and the induced classical measure, with exact and numerical comparisons.
Semester 10	Integration and validation of results, determination of the conditions of validity of the approximations, and identification of open problems. Preparation of manuscripts, final thesis writing, and defense.

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